CAMBRIDGE O LEVEL COMPUTER SCIENCE (2210 / 0478)

Paper 1: Computer Systems | Topic-Wise Question Bank (2024 – 2026 Series)

TOPIC 05: THE INTERNET AND CYBER SECURITY

Exam Sources: Cambridge 2210 / 0478 (May/June 2023, Oct/Nov 2023, May/June 2024)

Question 1 [2210/12/M/J/24 Q8 (2024 Series)] [Total: 7 marks]

(a) Explain the difference between an IP address and a MAC address. **[2 marks]**

(b) Describe how a Domain Name System (DNS) server translates a URL into an IP address. **[2 marks]**

(c) Describe the mechanism of a Distributed Denial of Service (DDoS) attack and state its effect on a web server. **[3 marks]**

✓ Marking Scheme / Model Answer:

(a) IP address: Logical/network address assigned by network; can change [1 mark]. MAC address: Physical hardware address hardcoded on NIC; permanent [1 mark].
(b) DNS: Browser sends URL request -> DNS server looks up domain in database -> retrieves matching IP address -> sends IP back to browser [2 marks].
(c) DDoS: Botnet of compromised computers floods target web server with overwhelming volume of requests simultaneously [2 marks]. Server becomes overwhelmed and crashes / is unavailable to legitimate users [1 mark].

Question 2 [2210/12/M/J/23 Q6 (2023 Series)] [Total: 6 marks]

(a) Explain why HTTPS is more secure than standard HTTP for online banking. **[2 marks]**

(b) Explain what cookies are and differentiate between session cookies and persistent cookies. **[3 marks]**

(c) State what is meant by digital currency / blockchain technology. **[1 mark]**

✓ Marking Scheme / Model Answer:

(a) HTTPS encrypts data using SSL/TLS protocols; prevents hackers from eavesdropping or intercepting confidential financial data [2 marks].
(b) Cookies: Small text files stored on user's computer by websites to store user preferences and session data [1 mark].
Session cookies: Deleted when browser is closed [1 mark].
Persistent cookies: Remain on user device until expiration date to remember login/settings [1 mark].
(c) Blockchain: Decentralized, distributed digital ledger that securely records transactions across a peer-to-peer network using cryptographic hashing [1 mark].

Question 3 [2210/12/O/N/23 Q5 (2023 Series)] [Total: 5 marks]

(a) Distinguish between the following types of malware:

(i) Computer Virus

(ii) Computer Worm

(iii) Trojan Horse **[3 marks]**

(b) Describe the role of a proxy server in network security. **[2 marks]**

✓ Marking Scheme / Model Answer:

(a) (i) Virus: Attaches to host files and requires human action/execution to spread [1 mark].
(ii) Worm: Self-replicating standalone malware that spreads across networks automatically without host files [1 mark].
(iii) Trojan Horse: Disguised as legitimate software to trick user into installing it, creating a backdoor [1 mark].
(b) Proxy server: Acts as intermediary between user and internet; hides user's real IP address; filters content; caches web pages for faster access [2 marks].